

ENDOMETRIUM AND MYOMETRIUM

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Introduction

- Both make up uterine wall
- Endometrium is the mucosal lining of the uterine cavity
- Myometrium is the smooth muscles underling the endometrium

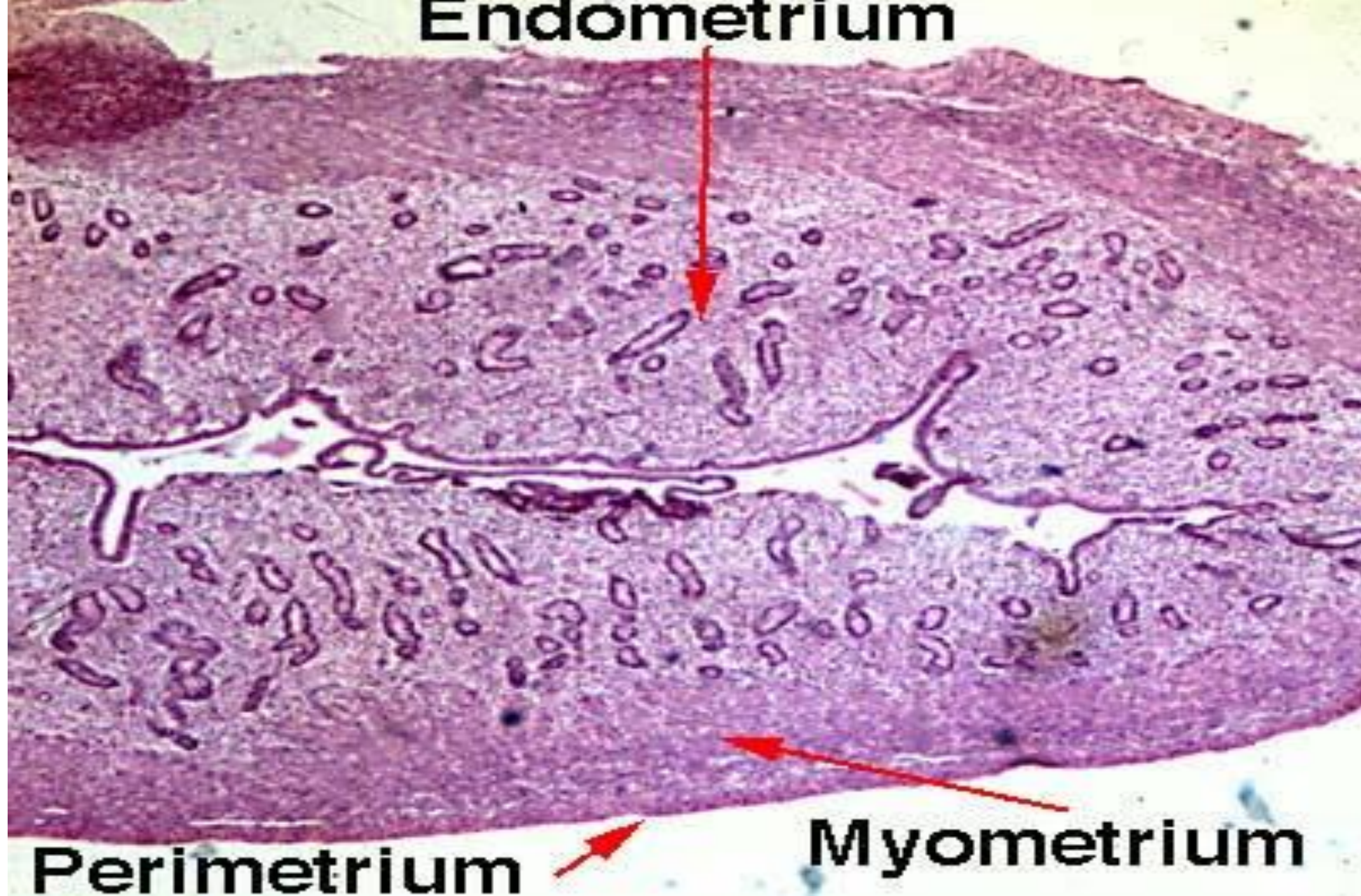
Endometrium



Myometrium



Perimetrium



Endometrium is hormonally sensitive

- Growth of the endometrium is estrogen driven
- Preparation of the endometrium for plantation is progesterone driven
- Shedding occurs with loss of progesterone support

• Grows  prepares  sheds

ASHERMAN SYNDROM

- Secondary amenorrhea due to loss of basalis (regenerative layer) and scarring
- Result of overaggressive dilation and curettage (D and C)
- Rare but possible result of d&c

Anovulatory cycle

- Lack of ovulation
- Results in estrogen driven proliferative phase without progesterone driven secretory phase
- Common cause of dysfunctional uterine bleeding, especially during menarche and menopause.

Acute endometritis

- Bacterial infection of endometrium
- Usually due to retained products of conception
- Presents as fever, abnormal uterine bleeding and pelvic pain

Chronic endometritis

- Chronic inflammation of endometrium
- Characterised by plasma cell
- Common cause include retained products of conception, chronic PID, IUD and TB
- Presents with abnormal uterine bleeding, pelvic pain and infertility

Endometrial polyp

- Hyperplastic protrusion of endometrium
- Presents as abnormal uterine bleeding
- Can arise as a side effect of tamoxifen

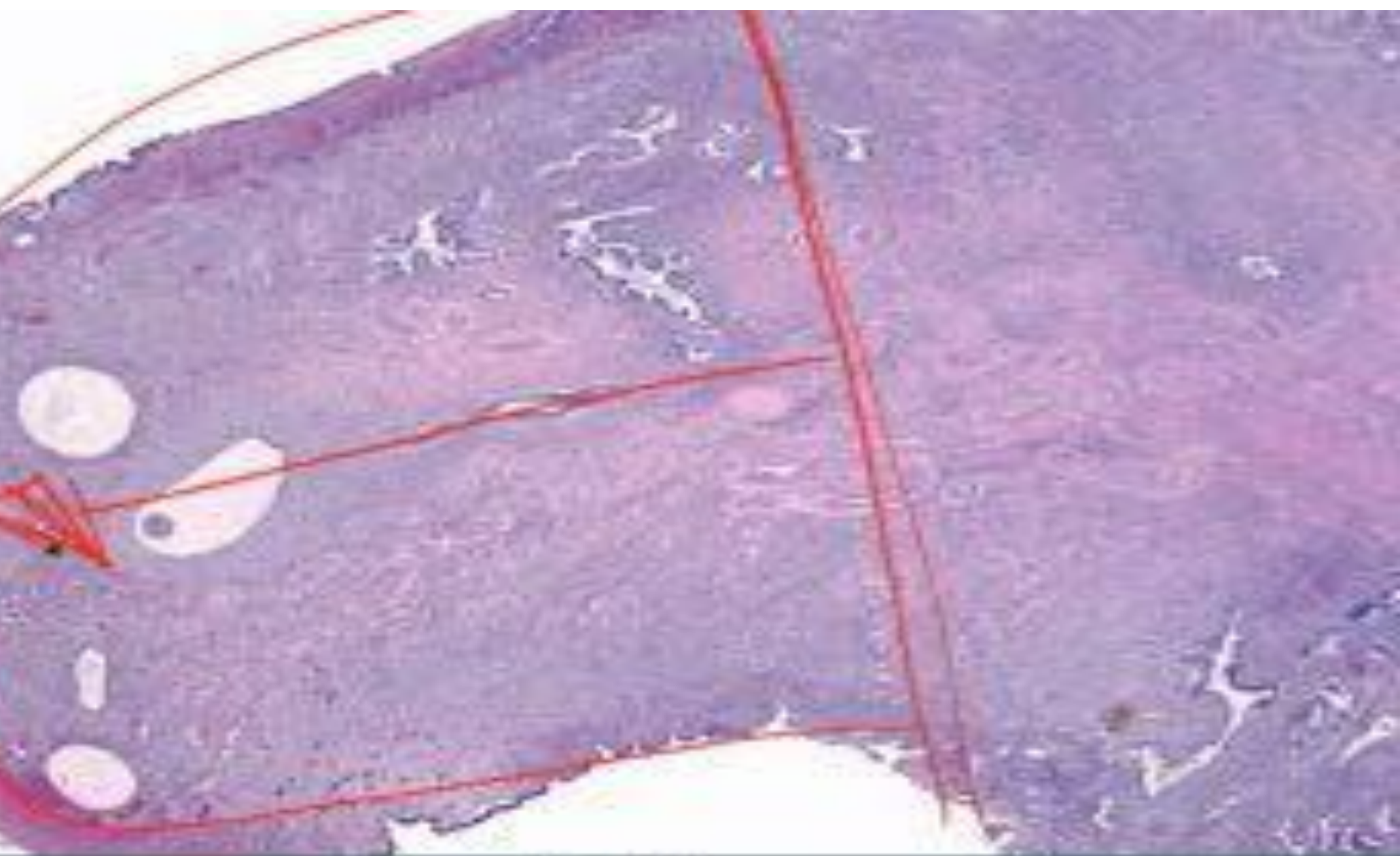


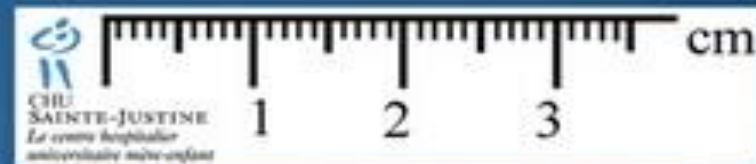
Fig. 17.6

endometriosis

- Endometrial glands and stroma outside the uterine endometrial lining
- Presents with dysmenorrhea and pelvic pain
- May cause infertility
- Causes of endometriosis
 - Metaplastic theory
 - Lymphatic dissemination theory

Common sites of involvement

- Ovary --- chocolate cyst
- Uterine ligaments – pelvic pain
- Pouch of Douglas – pain with defecation
- Bladder wall – pain with urination
- Bowel serosa – abdominal pain and adhesion
- Fallopian tube mucosa - scarring



Involvement of uterine myometrium is called adenomyosis

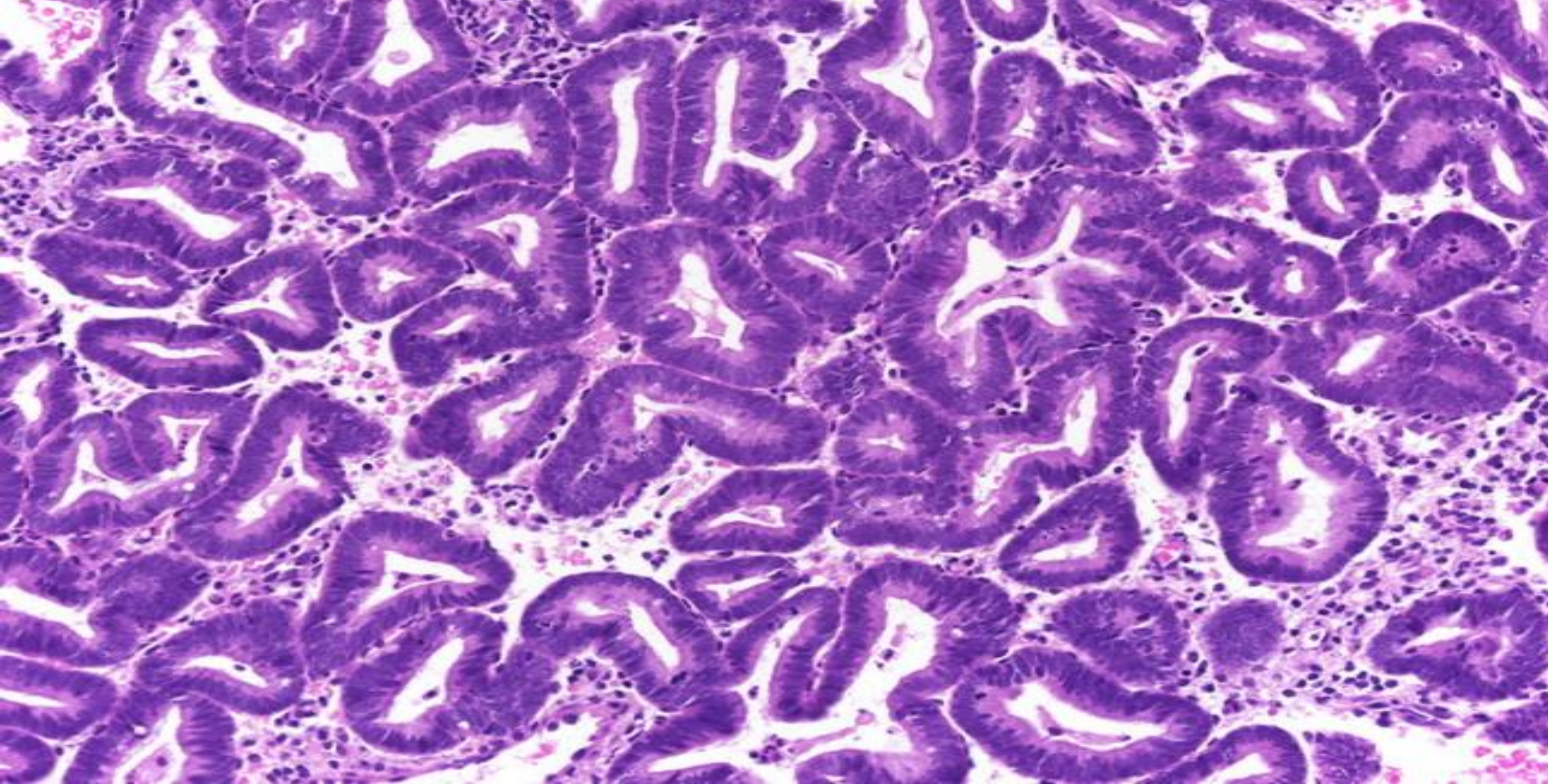
- Presence of endometrial cells and stromal in the myometrium or smooth muscle .
- Or presence of endometriosis in the myometrium

Increased risk of carcinoma at the site of endometriosis

- Especially the ovary

Endometrial hyperplasia

- Hyperplasia of endometrial glands relative to stroma
- Consequence of unopposed estrogen
- Presents as post menopausal uterine bleeding

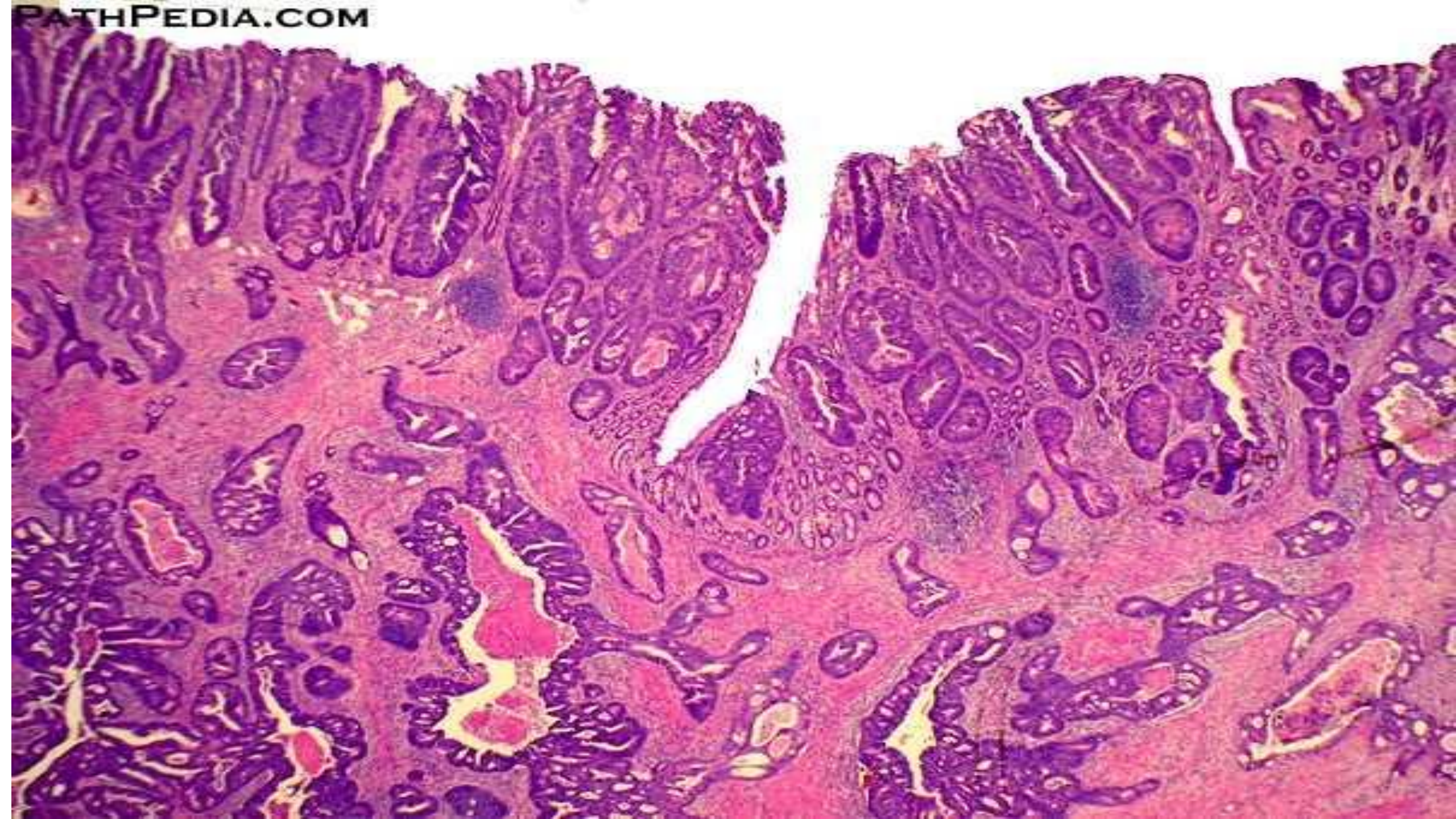


Classified histologically

- Based on architectural growth and cellular atypia
- Most important predictor for progression to carcinoma is cellular atypia

Endometrial carcinoma

- Malignant proliferation of endometrial glands
- Presents as postmenopausal bleeding
- Arises via two distinct pathways
 - Hyperplasia → carcinoma occurs in women 50 to 60 years
 - Sporadic → serous (papillary) occurs in women 70 and above



Hyperplasia pathway

- Carcinoma arises from endometrial hyperplasia
- Risk factors are related to estrogen exposure
- Average age is 60 years
- Histology is endometrioid

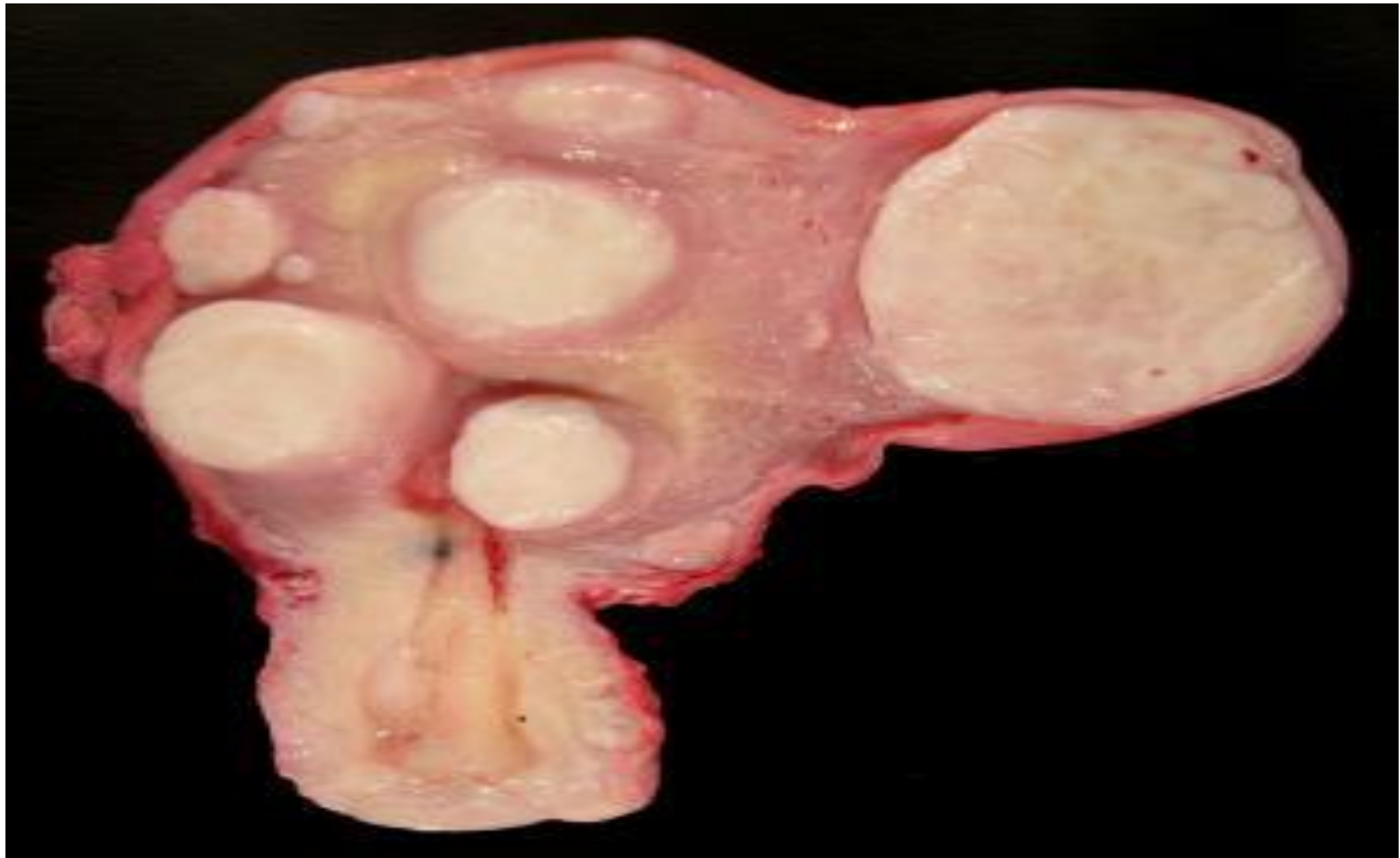
Sporadic pathway

- Carcinoma arises from atrophic endometrium
- No evident of precursor lesion
- Average age is 70 years
- Histology is usually serous
- Associated with p53 mutations
- Forms papillary structures and salmoma bodies

leiomyoma

- Benign proliferation of smooth muscle
- Arises from myometrium
- Related to estrogen exposure
- Multiple, well- defined white whorled masses

- Usually asymptomatic
- When present, symptoms include abnormal uterine bleeding, infertility and pelvic mass



leiomyosarcoma

- Malignant proliferation of smooth muscle
- It arises from myometrium
- Arises denovo
- Usually seen in postmenopausal women
- Single lesion with necrosis and haemorrhage
- Necrosis, mitotic activities and cellular atypia

